Solving Systems of Two Linear Equations by Graphing Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. $x = 2$, $y = 4$
2. $x = 4$, $y = 5$
3. $x = 3$, $y = 1$
4. $x = 3$, $y = 2$
5. $x = 2$, $y = 2$
6. $x = 2$, $y = 30$
7. $x = 4$, $y = 0$
8. —
9. $x = 2$, $y = 14$
10. $x = 3$, $y = 4$
11. $x = 0$, $y = 4$
12. $x = 8$, $y = 4$

Curriculum

Level: Grade 8 / Pre-Algebra
8.EE.C.8 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

1. $y = x + 2$ **and** $y = -x + 6$**.** The first line starts at 2 on the y-axis and rises 1 for every 1 across. The second starts at 6 and falls 1 for every 1 across. Plotting both, the crossing point sits two to the right of the y-axis. Check by substituting: $4 = 2 + 2$ is true, and $4 = -2 + 6$ is true, so the point lies on both lines. $\boxed{(2,\ 4)}$

2. $y = 2x - 3$ **and** $y = x + 1$**.** Line one starts at -3 and climbs steeply (up 2, across 1); line two starts at 1 and climbs gently. The steeper line begins lower and catches up. Substituting the crossing point: $5 = 2(4) - 3$ is true and $5 = 4 + 1$ is true. $\boxed{(4,\ 5)}$

3. $y = -2x + 7$ **and** $y = x - 2$**.** The first slope is negative, so that line falls; the second is positive, so it rises. A falling line and a rising line always cross exactly once, which is why this system has one solution before you graph anything. Check: $1 = -2(3) + 7$ and $1 = 3 - 2$ are both true. $\boxed{(3,\ 1)}$

4. $x + y = 5$ **and** $x - y = 1$ **(standard form).** Use intercepts. For $x + y = 5$: setting $x = 0$ gives $(0, 5)$, and setting $y = 0$ gives $(5, 0)$. For $x - y = 1$: setting $x = 0$ gives $(0, -1)$, and setting $y = 0$ gives $(1, 0)$. Draw each line through its two intercepts. Check: $3 + 2 = 5$ and $3 - 2 = 1$. $\boxed{(3,\ 2)}$

5. $y = \frac{1}{2}x + 1$ **and** $y = -x + 4$. A slope of $\frac{1}{2}$ means up 1, across 2 — take two steps of that from $(0, 1)$ rather than guessing. The second line falls one for one from $(0, 4)$. Check: $2 = \frac{1}{2}(2) + 1$ and $2 = -2 + 4$. $\boxed{(2,\ 2)}$

6. Club A: $y = 5x + 20$**; Club B:** $y = 10x + 10$**.** Club A starts higher on the vertical axis (the bigger joining fee) but rises more slowly, so Club B catches up. Plot $(0, 20)$ and step up 5 per month; plot $(0, 10)$ and step up 10 per month. The lines meet after two months, when both clubs have cost the same amount. Check: $5(2) + 20 = 30$ and $10(2) + 10 = 30$. $\boxed{(2,\ 30)}$

Interpretation to expect in words: at two months both cost 30 dollars; before that Club B is cheaper, after that Club A is cheaper.

7. $2x + y = 8$ **and** $x - y = 4$ **(standard form).** Intercepts again. For $2x + y = 8$: $(0, 8)$ and $(4, 0)$. For $x - y = 4$: $(0, -4)$ and $(4, 0)$. Both lines pass through the same x-intercept, so the crossing point sits on the x-axis and its y-coordinate is zero. Check: $2(4) + 0 = 8$ and $4 - 0 = 4$. $\boxed{(4,\ 0)}$

8. $y = 3x + 1$ **and** $y = 3x - 2$**.** Both slopes are 3, so the two lines climb at exactly the same rate; they start at different heights (1 and -2) and stay a constant distance apart forever. Parallel lines never cross, so the graph shows no crossing point at all. Algebra agrees: setting them equal gives $1 = -2$, which is impossible. $\boxed{\text{no solution}}$

9. Plant A: $y = 4x + 6$**; Plant B:** $y = 6x + 2$**.** Plant A is taller at the start but grows more slowly, so the steeper line overtakes it. Plot $(0, 6)$ rising 4 per week and $(0, 2)$ rising 6 per week. Check: $4(2) + 6 = 14$ and $6(2) + 2 = 14$. $\boxed{(2,\ 14)}$

Interpretation: after two weeks both plants are 14 cm tall; after that Plant B is the taller one.

10. $3x - y = 5$ **and** $x + y = 7$**.** Rearranging is the safer route here because neither line has friendly intercepts on the printed grid. From $3x - y = 5$, add y and subtract 5: $y = 3x - 5$. From $x + y = 7$: $y = -x + 7$. Now graph from the intercepts and slopes. Check: $3(3) - 4 = 5$ and $3 + 4 = 7$. $\boxed{(3,\ 4)}$

11. $2x + 2y = 8$ **and** $y = -x + 4$**.** Divide the first equation by 2: $x + y = 4$, which rearranges to $y = -x + 4$ — exactly the second equation. The two graphs are the same line drawn twice, so every point on it satisfies both equations and the system has infinitely many solutions. $\boxed{y = -x + 4}$

Grading note: accept “infinitely many solutions”, “all points on the line”, or a named example such as $(0, 4)$ or $(4, 0)$. The graph shows one line, not two. This is the one problem on the sheet whose answer is a whole line rather than a single point, so it is flagged for your review rather than machine-checked as a single pair.

12. Tall candle: $y = -2x + 20$**; short candle:** $y = -x + 12$**.** Both slopes are negative because both candles shrink. The tall candle starts higher but burns twice as fast, so it must fall through the short candle’s height at some point. Setting the heights equal: $-2x + 20 = -x + 12$ gives $8 = x$, and then $y = -8 + 12 = 4$. Check on the graph: at eight hours both curves are at height 4 cm. $\boxed{(8,\ 4)}$

Which is taller before then: the tall candle — its line lies above the other one for every hour before the crossing point, and below it afterwards.